Horizon-dependent Benefits and Reliability Limits of Mixture-SDE Movement Forecasting

Public review revision, 14 October 2026; authorship unconfirmed

Abstract

A movement forecast can improve its distributional score without locating the target more accurately or providing reliable uncertainty regions. We examine these distinctions in an existing evaluation of fitted stochastic movement predictors applied to historical GPX position recordings with unverified carrier identities and transport modes, using 46 retrospectively selected recording-hash blocks, five simulation seeds and 512 particles per ensemble. At nominal 1 and 5 minutes, deterministic constant-speed extrapolation has lower energy score (ES) than the heading-stratified mixture SDE (HSM-SDE); at 15 and 30 minutes the ordering reverses. Averaging the four targets gives ES 477.85 m for HSM-SDE and 824.73 m for constant-velocity extrapolation (CV). A residual-direction mixture variant (RDM-SDE) has lower average ES and final mean-position error, but its paired ES interval includes zero. Observation intervals of 120 and 300 saved seconds have lower average ES than the 60 s reference, whereas the examined training/adaptation variants show small ES changes without a confirmed added decrease. The 300 s variant reduces average ES by 23.40 m but increases final mean-position error by 9.64 m. HSM-SDE's nominal 90% prediction disk covers only 46.52% of 30-minute targets. These are conditional descriptive findings: participant independence and physical clock provenance remain unverified, and the cohort was selected using future-route map support. Required forecast failures prevent registered terrain-family inference; a separate post-hoc 43-block comparison shows higher mean ES for all-terrain than its history-only base and horizon-dependent single-group trade-offs. The study identifies conditional long-horizon utility and design trade-offs, not the necessity of SDE complexity or a confirmed structural winner. It demonstrates an executable probabilistic forecasting procedure, but does not establish reliable long-horizon regions or superiority over a simple probabilistic baseline.

1 Introduction

Movement prediction is often summarized by a single accuracy measure, yet three distinct tasks are involved: assigning a distribution to future positions, locating a representative position, and describing a region that contains the realized target at its stated frequency. A gain on one task need not imply a gain on the others. This distinction matters when a probabilistic forecast is interpreted as a reliable search region.

Continuous-time correlated random walks model movement persistence and irregular observations [1]; switching state-space models represent latent regimes [2]. Step-selection models link movement to environmental selection [3, 4]. Learned trajectory predictors [5, 6], neural SDE generators [7] and interpretable stochastic system identification [8] offer additional modelling approaches. These works provide context, not performance baselines executed here. Our empirical subject is future position in selected geographic recordings; the modelling object is a conditional distribution over those positions. A stochastic position model represents uncertainty through a fitted drift,

residual diffusion and latent components. The study evaluates an existing, limited collection of these fitted predictors.

Proper scoring rules evaluate predictive distributions against realized targets [9, 10]; they do not certify coverage of a particular prediction region. We therefore ask how distributional score, error of the predictive mean and marginal disk coverage change across four horizons. Our primary concern is the utility and reliability of longer-horizon forecasts, with shorter horizons providing a contrast. The argument asks when CV or HSM-SDE performs better; how RDM-SDE compares with HSM-SDE; how observation interval changes score and location; whether training/adaptation strategies add a benefit; and whether the resulting long-horizon prediction disks can be trusted. Score–location disagreement is used to interpret these choices, not treated as a separate list of unrelated experimental outcomes.

This focus is an interpretation of existing results, not a newly registered hypothesis test. The original questions remain explicit: RQ1 asks whether the registered changes in mode structure, observation interval, fitting/adaptation and numerical prediction yield a practically useful distribution-score change relative to each group's specified reference procedure. RQ2 asks whether selected terrain groups add predictive value beyond a fitted history baseline, through overall, leave-one-group-out and leave-one-group-in comparisons. Those registered questions and inference rules are retained. The questions organising the present paper are descriptive readings of the existing evidence, not additional registered tests. We first define the recordings and forecast task, explain the model construction and evaluation criteria, then move from horizon-specific utility through design choices to region reliability. Terrain efficacy is reported separately as an additional descriptive analysis with explicit complete-case selection; its unavailable registered inference is not treated as absence of results or as a demonstrated benefit of the ordinary SDE.

2 Data and predictive procedures

2.1 Research object and forecasting task

The empirical objects are timestamped geographic position recordings from a historical GPX collection. Each recording traces the movement of an unidentified recording carrier. The retained archive does not establish whether a carrier is walking, cycling or travelling by another means, or whether two recordings belong to the same person. We therefore study position changes in the selected recordings, rather than the behaviour of a defined participant population. A cleaned segment is a derived portion of a recording; a window splits that portion into an observed prefix and future targets. Multiple windows can belong to one recording-hash block. Only one primary origin per selected block enters the main evaluation.

For an origin with saved time t_0, let $\mathcal{H}_0$ contain the allowed observed position/time prefix, ending at the current position x_0. Its last three pairs initialize the retained motion-history buffer. The task is to predict future positions conditional on this visible information and on fixed development-fitted parameters $\widehat{\xi}$. The method's internal east/north frame is anchored at the first visible position of that whole prefix, which need not be the first of the last three buffered points. Terrain prediction and evaluation instead use a frame anchored at the last visible position, the forecast origin. Method outputs are converted through geographic coordinates into that common scoring frame; the distributions in Equation (1) are expressed there. The method also has the retained clock and its solar proxy; the terrain branch additionally queries fixed map fields at simulated positions. True future positions are reserved for selection and evaluation and do not update a forecast. With actual

matched elapsed times $u_1, \ldots, u_4$, the reported predictive object is

$$\widehat{P}_j = \mathcal{L}_{\widehat{\xi}}(X_{t_0+u_j} \mid \mathcal{H}_0), \qquad \widehat{P}_{N,j} = \frac{1}{N}\sum_{p=1}^{N} \delta_{X_j^{(p)}}, \quad j = 1, \ldots, 4. \tag{1}$$

Here δ_x is a point mass at x, and each simulated particle p has a position at all four targets. The ensemble approximates a model-defined distribution; it is not a sample of people. Its mean gives a representative location, while its radial quantiles define prediction disks. These summaries answer different questions about the same forecast. No separate measurement-error model is fitted: the saved target position is used as the evaluation reference, although its physical accuracy is not independently verified.

Development windows supply parameter estimates; final-evaluation windows supply forecast origins and observed outcomes. The main analysis asks how well those fixed predictors describe these outcomes as the saved forecast horizon increases. It does not estimate a causal law of human movement or the probability of finding a person in an operational search.

2.2 Evaluation population and clock

The task is single-recording position forecasting from a visible trajectory prefix. The retained data pipeline identifies its source as a historical OpenStreetMap GPX harvest, not the separate GeoLife cohort used in earlier work. User identities, transport-mode labels and original GPX segment boundaries are not preserved; these records cannot be treated as verified pedestrian participants. R1-derived segments supply midpoint history/target windows. The frozen recording-hash assignment separates development and evaluation: qualified development contains 404 outer-training windows (328 training and 76 adaptation) and 81 validation windows. Terrain fitting uses one first-future transition per outer-training window. Exact identity intersections across splits are zero, which does not establish participant or near-route independence. The scientific supplement details this assignment.

The released final-evaluation split contains 12,370 windows from 1,094 recording-hash blocks. Temporal eligibility retains 127 windows/89 blocks, joint feature validity retains 106/73, and the fixed primary-origin rule selects 46/46 (Figure 1). A block identifies a saved recording, not a verified independent participant. The selected cohort spans 12 countries and 27 source labels; 33/46 blocks are from Great Britain, Belgium or Luxembourg. It is not a representative population sample.

Future observations are used to select this cohort and to score forecasts, but are not inputs to prediction: online map queries use simulated positions. This distinction prevents direct target leakage during rollout without removing the retrospective selection restriction. The evaluation data had previously been inspected; the full influence of that inspection on design choices is not recoverable. This is not a fresh blinded holdout.

Targets are the actual observations matched to nominal 1, 5, 15 and 30 minutes, with elapsed-time ranges 49–67, 294–305, 889–910 and 1785–1814 saved seconds. No target is interpolated to an exact nominal time. Physical acquisition-time and UTC provenance are unverified. An absolute offset affects the calendar input; a relative-scale distortion also affects rates, diffusion units and the physical interpretation of the horizons. All minute labels below refer to the saved clock.

2.3 Three-component SDEs: design, training and forecasting

2.3.1 Baselines and shared structure

We study two procedures within one three-component stochastic differential equation (SDE) framework: the heading-stratified mixture stochastic differential equation (HSM-SDE), and the residual-

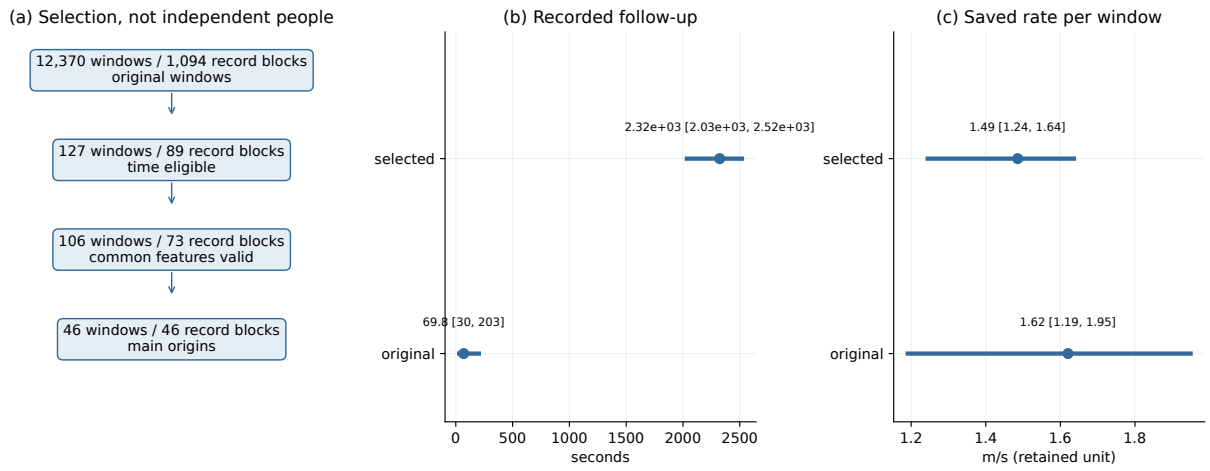


Figure 1: Evaluation selection and population descriptors. Counts distinguish windows from recording blocks; duration and speed panels show saved medians and interquartile ranges. Eligibility includes map validity along the observed future route, which is unavailable at deployment from the prefix alone. Selection exclusions are not forecast failures.

direction mixture stochastic differential equation (RDM-SDE). HSM groups fitting segments by heading; RDM constructs transition groups once from residual directions, rather than fitting a mixture likelihood by expectation–maximisation. They share a three-component SDE equation template but differ in grouping-based fitting and forecast-time component persistence. They are two complete procedures within one model family, not unrelated architectures or merely two training algorithms. The names are paper-specific descriptive shorthand, not a mixture-distribution versus SDE comparison. The deterministic reference is constant-velocity extrapolation (CV). The main paper and current figures/tables use these reader-facing names; numbers after HSM distinguish registered reference procedures, not new architectures. The audit record alone preserves the original identifier crosswalk; no saved identity or comparison control changes.

Table 1: Training and forecasting mechanisms of the three ordinary procedures. Components are fitted movement rules, not validated behaviour classes. HSM versus RDM changes both mechanism columns.

Program / role	Training: obtaining the rules	Forecasting: using the rules
CV / deterministic baseline	No development-set fitting; estimate velocity from the permitted prefix	Extrapolate at that velocity; no component draw
HSM / ordinary reference	Sort segment net headings into three groups; fit drift and diffusion in each	Draw a component per particle at the origin; keep it throughout the path
RDM / family variant	Fit a pooled regression; group transition residual directions; refit within each group	Draw at the origin; independently redraw every τ, allowing the same component

Deterministic baseline CV. For the 46 primary origins, the origin-state buffer retains the last three observed positions and their strictly increasing saved times. The origin is the last point. CV extrapolates $\widehat{x}^{\mathrm{CV}}(u) = x_0 + u\widehat{v}_0$, where $\widehat{v}_0$ is the secant from the first to last of those three points, divided by their saved elapsed time; no future position or future speed is used. These spans range from 2 to 61 saved seconds (median 7). CV fits no development-set parameters and adds no random

diffusion. Ordinary HSM-01 is the main method-family reference, not the separate terrain branch's base, defined with all-terrain below.

Shared stochastic position model. In saved local east/north coordinates $X = (X_e, X_n)^\top$ (metres), both use

$$dX_t = \Theta_{m_t}^\top \psi(X_t, t)\, dt + L_{m_t}\, dW_t, \quad \psi = (1, X_e, X_n, S)^\top, \quad L_m L_m^\top = Q_m = \tau R_m. \tag{2}$$

The first visible position anchors the coordinate frame; forecasting starts at the last permitted prefix position. S is a solar/calendar proxy computed from the retained absolute tick [15], not a validated daylight mechanism. R_m is a centred fitted displacement-rate residual covariance; its conversion uses observation interval τ, not integration step h. Given frame, position, clock and random stream, neither ordinary HSM nor RDM uses initial speed or recent heading in its drift, diffusion or mode probabilities. HSM-SDE and CV thus receive the same permitted prefix but use its movement information differently; training heading labels are not a forecast-time heading classifier.

What the stochastic equation represents. $X_t \in \mathbb{R}^2$ is position and $m_t \in \{1, 2, 3\}$ is an unobserved model component. $\Theta_m \in \mathbb{R}^{4\times 2}$ maps the intercept, two position coordinates and solar proxy to an east/north drift $b_m = \Theta_m^\top \psi$. This drift is a conditional displacement rate, measured in metres per saved second. W_t is a two-dimensional standard Wiener process with independent increments of covariance ΔI_2 over duration Δ; L_m maps them to position noise. Thus $Q_m = L_m L_m^\top$ is a diffusion covariance in square metres per saved second, and L_m has units of metres per square root of a saved second. The deterministic and random parts respectively represent a fitted average trend and residual variation, not a separately observed intention or velocity process.

For frozen drift and diffusion, the local increment has the illustrative law

$$X_{t+\Delta} - X_t \mid (X_t, m_t = m) \;\approx\; \mathcal{N}(b_m \Delta, Q_m \Delta)\,. \tag{3}$$

The mean displacement grows with Δ, while its random standard deviation grows with $\sqrt{\Delta}$. A sampled component adds uncertainty about which fitted trend applies. For constant component drifts and a segment-fixed mode, the corresponding illustrative mixture covariance is $u \sum_m \pi_m Q_m + u^2 \operatorname{Cov}_\pi(b_m)$: within-component diffusion and between-component drift variation need not grow at the same rate. HSM has position- and solar-dependent drift, so this expression explains the ingredients rather than gives its actual multi-step variance or proves calibrated coverage.

2.3.2 Training groups, estimation and validation selection

HSM: hard grouping by segment heading. Development segments are resampled at the registered observation interval τ; an incomplete terminal interval is removed from fitting, while scoring observations remain unchanged. At $\tau = 60$ saved seconds, the 328 training, 76 adaptation and 81 validation windows supply 26,009, 5,762 and 6,149 complete transitions, respectively. A transition target is the two-dimensional displacement rate $y_i = (x_{i+1} - x_i)/\tau$. Within each fitting batch, segments are sorted by their net heading and divided into three groups approximately balanced in segment count; all transitions from a segment inherit its group. This is within-batch ordering, not three fixed geographic angle sectors; training and adaptation batches are grouped separately. The groups identify regression components, not externally validated walking, turning

or transport states. For group m, with feature matrix D_m and rate matrix Y_m, the base estimates are

$$\widehat{\Theta}_m = \underset{\Theta}{\operatorname{argmin}} \left\{ \tfrac{1}{2}\|D_m\Theta - Y_m\|_F^2 + \tfrac{\rho}{2}\|\Theta\|_F^2 \right\}, \qquad \rho = 10^{-6}, \tag{4}$$

$$\widehat{R}_m = \frac{1}{n_m - 1}\sum_{i\in m}(e_i - \bar{e}_m)(e_i - \bar{e}_m)^\top + 10^{-8} I_2, \quad e_i = y_i - \widehat{\Theta}_m^\top \psi_i. \tag{5}$$

The Frobenius norm sums squared entries; all coefficients, including the intercept, are penalised. n_m is the number of transitions in group m, and its base mixture weight is $n_m / \sum_k n_k$. R_m measures centred residual variation in squared displacement-rate units. The small diagonal term stabilises the covariance calculation; it does not calibrate prediction regions. These are regression estimates, not continuous-time path-likelihood estimates.

RDM: hard grouping by transition residual direction and refitting. A pooled ridge regression with the same features first estimates $\widehat{\Theta}_{\text{pool}}$ from the whole batch. With residuals $r_i = y_i - \widehat{\Theta}_{\text{pool}}^\top \psi_i$, one-time labels are

$$g_i = \arg\max\{r_{i,e},\ r_{i,n},\ -r_{i,e} - r_{i,n}\}. \tag{6}$$

The three score indices define the groups; transitions within one segment can enter different groups. Each group is then refitted to the original rate targets y_i using Equations (4)–(5), obtaining a complete drift regression and residual covariance. It does not fit a residual correction on top of the pooled drift: the pooled fit only produces labels and is not retained as a common drift. This is deterministic hard grouping, without EM iterations or soft responsibilities. Thus base fitting can indeed be described as dividing data into three groups and independently fitting one regression component per group. Features and ridge strength are shared, not the group-specific fitted coefficients.

Default training/adaptation procedure shared by both. HSM-01 and RDM fit training and adaptation batches separately, then combine corresponding coefficients, covariances and proportions with weights 0.35 and 0.65. This is neither a single fit on pooled training/adaptation data nor gradient fine-tuning from the training parameters. Validation selects a covariance scale $s \in \{0.55, 0.75, 1, 1.3, 1.7\}$ with drift held fixed. The saved R_m already includes this s^2 scaling. Its continuous-time embedding uses $Q_m = \tau R_m$: at a constant-coefficient duration τ, $Q_m\tau = R_m\tau^2$ matches the displacement covariance implied by the rate regression. This explains the time factor; it does not establish equivalence to maximum-likelihood estimation of an SDE. The chosen parameters remain fixed across evaluation origins and seeds; the final component probabilities include the training/adaptation blend. HSM's saved component probabilities are 0.3350, 0.3302 and 0.3348, rather than a posterior re-estimated from each new prefix. HSM/RDM identify these constructions and forecast procedures. Scratch, Reptile, drift-only and two-step separately modify training/adaptation strategies or parameter-update scope; they are not training-method labels at the same level as the two procedure names.

The validation selection proxy is a one-step rate-residual Gaussian using the unweighted average mode covariance, whereas evaluation scores a multi-mode, multi-step position distribution. The RDM variant groups training residuals once by $\arg\max(r_e, r_n, -r_e - r_n)$ but uses segment-heading labels for validation components: these labels are not aligned. Independently heading-sorted batches are also mixed by mode index in the transfer procedures [12]. These are implementation limitations, not evidence that the corresponding modelling ideas fail.

2.3.3 Forecasting and component updates

Evaluation uses $N = 512$ particles, five fixed seeds and maximum integration step $h = 5$ saved seconds. HSM uses conditional Gaussian moment propagation, evaluating nonlinear conditions at the conditional mean; this is not an exact solution of a globally nonlinear SDE [11]. Four targets come from a continuing joint path, not independently sampled marginals. Observation-interval variants change fitting data, $Q = \tau R$ and update cadence together while retaining the integration cap. In particular, dt300 and dt600 mean observation intervals of 300 and 600 saved seconds, not integration steps. Identical-parameter aliases and multi-factor changes make the 28 saved identities procedures rather than 28 independent algorithms. All 21 registered primary contrasts retain their specified reference procedures and reference codes.

Component schedules before position propagation. Each particle starts at its internal-frame origin position X_0 and samples a component from its procedure's own final saved weights π. HSM keeps that component throughout its path. RDM independently redraws at the end of every reference interval $\tau = 60$ saved seconds for the next interval, until the forecast endpoint. If M_k denotes the component in interval k, then

$$\Pr(M_{k+1} = j \mid M_k = i) = \begin{cases} \mathbf{1}\{j = i\}, & \text{HSM}, \\ \pi_j, & \text{RDM}. \end{cases} \tag{7}$$

RDM is not a cyclic $1 \to 2 \to 3$ rotation or a mandatory change: it can draw the same component again. Redrawing does not depend on position, speed, recent heading or behaviour recognition. Its component clock differs from the maximum integration step $h = 5$ seconds: a 60-second interval typically contains twelve 5-second numerical steps, not twelve component redraws. For illustration, either procedure can produce component sequence $2, 2, 2$; RDM, but not HSM, can produce $2, 1, 3$. These indices illustrate rules, not saved experimental trajectories. Thus the comparison changes training groups, fitted parameters and forecast-time persistence together; it cannot isolate the benefit of a grouping or redrawing mechanism.

Position generation conditional on a component schedule. On each event interval $\Delta \le 5$ saved seconds, write the locally affine drift as $\mathsf{A}_m x + a_n$, freezing the non-position feature at the conditional mean. A_m is the transpose of the two position rows of Θ_m; a_n contains the intercept and frozen solar term. HSM and RDM both use the registered FP/split procedure: an exact affine half-drift, an additive noise increment, then another half-drift. Its matrices and conditional-moment update are

$$H = e^{\Delta \mathsf{A}_m/2}, \qquad J_{1/2} = \int_0^{\Delta/2} e^{v\mathsf{A}_m}\, dv,$$

$$F = H^2, \qquad J = (I + H)J_{1/2}, \qquad \Omega = H(\Delta Q_m)H^\top, \tag{8}$$

$$\mu_{n+1} = F\mu_n + Ja_n, \qquad C_{n+1} = FC_nF^\top + \Omega, \tag{9}$$

$$X_{n+1}^{(p)} = FX_n^{(p)} + Ja_n + Gz_{n,p}, \quad GG^\top = \Omega, \quad z_{n,p} \sim \mathcal{N}(0, I_2). \tag{10}$$

Initially $\mu_0 = X_0$ and $C_0 = 0$. The same noisy path supplies its positions at all four actual target times. The drift substeps are exact for frozen affine coefficients; the split covariance generally differs from the affine-exact integral $\int_0^\Delta e^{v\mathsf{A}_m} Q_m e^{v\mathsf{A}_m^\top}\, dv$. Evaluating the solar feature at the conditional mean also approximates the nonlinear full model. In particular, "FP" in the saved procedure denotes conditional Gaussian moments, not a spatial Fokker–Planck PDE solution. Numerical

variants use Euler–Maruyama or affine-exact reference kernels, and MC/CRN variants evaluate nonlinear conditions along individual paths. Their complete definitions are in the scientific supplement. This construction produces the empirical measures in Equation (1); no subsequent radius calibration is part of the evaluated forecast.

2.4 Map factors: roads, rivers, land cover and surface geometry

The separate terrain branch augments a fitted three-position history drift with independently fitted residual conditioners and diffusion for ten configurations. Its fitted mechanism is specified below. The ordinary HSM/RDM variable S is only the solar/calendar proxy, not these map factors. Here the map input is the explicitly expanded $\widetilde{\phi}_g$ below; the two conditioning inputs must not be conflated. This branch fits its own history drift and residual conditioner; it does not reuse HSM/RDM's three sets of component parameters or add maps to their S.

The terrain model's inputs and fitted mechanism. Here X_t uses the forecast-origin-anchored east/north frame. Let Z_t be a buffer of up to three visible or generated positions and times. Its first-to-last secant gives v_t, its normalized direction gives h_t, and η_t indicates whether that direction is valid. The history design is $r(Z_t) = (1, v_{t,e}, v_{t,n}, \|v_t\|, h_{t,e}, h_{t,n}, \eta_t)^\top$. The terrain drift and position dynamics are

$$b_g(X_t, Z_t) = B_0^\top r(Z_t) + \theta_g^\top \widetilde{\phi}_g(X_t, Z_t), \qquad dX_t = b_g(X_t, Z_t)\,dt + L_g\,dW_t. \tag{11}$$

$\widetilde{\phi}_g$ appends an intercept to the selected map/history feature vector, which contains encoded numerical values and validity masks. The road and river inputs describe nearest-feature distances and directions; land cover uses grouped indicators; surface inputs describe slope/orientation; registered interactions combine selected map and history quantities. These are fixed engineering representations. Their coefficients are learned from displacement rates, not constrained by a law of road attraction or river avoidance.

Selected factors, not generic "water and roads". The road group uses retained Overture road lines. Its query excludes bridleway, cycleway, footway, path, pedestrian, steps and track classes; it is not a complete pedestrian network. The river group uses HydroRIVERS river lines, not all water surfaces, lake shores, fordability or bridge access. The cover group (saved key worldcover) uses WorldCover pixels; the surface group uses slope, aspect and normals from the retained DEM rules. Absolute elevation is a derivation source, not a selected direct conditioner column. Other path, rail, lake-shore distance, curvature and roughness representations are not selected in these ten configurations.

Exactly which distances, vectors and history products enter. At query position X, d_g is the distance in metres to the nearest selected line point, and $u_g = (u_{g,e}, u_{g,n})$ points from the query towards that point, for $g \in \{\text{road}, \text{river}\}$. The provider chooses the nearest point in its local metric projection and reports its spherical distance. Both direction components, log-distance and their product enter:

$$\ell_g = \log(1 + d_g/(1\,\text{m})), \qquad c_g = \ell_g u_g. \tag{12}$$

The vector is not a line tangent or river-flow direction. The two product columns are not a hard-coded road attraction or river repulsion law. No direct $u_{\text{river}}^\top h$, $u_{\text{river}}^\top v$ or river-distance times historical-speed input is selected. History velocity still enters the shared drift; generated-position feedback also couples motion and maps indirectly.

The four cover indicators w are built (50), vegetated (10,20,30,40,95,100), water/wetland (80,90), and bare/snow/other finite valid codes. Missing data are not valid "other". The surface vector is $s = (\beta, \sin\alpha, \cos\alpha, n_e, n_n, n_u)$: slope in radians, downhill aspect clockwise from north, and the three-dimensional upward unit normal. Write $a = -(\sin\alpha, \cos\alpha)$ for uphill and $q = (-a_n, a_e)$ for the chosen signed contour direction. The four selected compositions are exactly

$$c_{\text{road}} = \ell_{\text{road}} u_{\text{road}}, \quad c_{\text{river}} = \ell_{\text{river}} u_{\text{river}}, \quad c_{\text{cover}} = \ell_{\text{road}} w, \quad c_{\text{surface}} = (h^\top a, h^\top q). \tag{13}$$

Only the last composition explicitly combines terrain with history direction; it includes no slope- or speed-magnitude multiplier and is not all twelve surface-by-history products. Reversing q reverses the second projection. Flat-surface aspect is recorded as zero, not evidence of perpendicular travel.

Table 2: Named factors and actual numerical inputs. Dimensions exclude masks; each numerical column has its own validity indicator. Eighteen base columns plus ten composition columns give 28 numerical columns in the all-group configuration.

Group / source	Base factors (numerical dimension)	Dependent compositions
history / causal buffer	East/north unit history direction h (2)	With surface: $(h^\top a, h^\top q)$ (2)
road / road lines	Log-distance ℓ_{road} (1); towards-nearest-road direction u_{road} (2)	$\ell_{\text{road}} u_{\text{road}}$ (2); with cover: $\ell_{\text{road}} w$ (4)
river / river lines	Log-distance ℓ_{river} (1); towards-nearest-river direction u_{river} (2)	$\ell_{\text{river}} u_{\text{river}}$ (2); no direct history-velocity product
cover / land cover	Four grouped WorldCover indicators w (4)	With road: $\ell_{\text{road}} w$ (4)
surface / DEM	Slope, two downslope-aspect components, three normal components s (6)	With history: two signed directional projections (2)

Base, all, LOO and LIO are configuration operations, not factors. Let B be the base history configuration and A the all-terrain configuration. LOO means leave-one-group-out, $A - g$; LIO means leave-one-group-in above the history base, $B + g$, not a history-free single-factor model. Table 3 links these operations to result labels. Every configuration retains history. Removing a group removes all dependent compositions, followed by refitting its conditioner and diffusion.

The cover-road composition is removed with either road or cover; the two removal losses cannot be summed as independent factor contributions. Cover alone has no road distance and hence no such composition. LOO and LIO compare refitted predictors in different feature contexts, not input-zeroing of one fitted model or identified causal effects of distance, direction or an individual cover subclass.

Table 3: The ten configurations and retained factors/compositions. K counts conditioner numerical columns; input width is $2K$ including masks, with one additional fitted intercept. The separate seven-column history drift is not included in K. Results use B, $A-g$, $B+g$; original keys remain in the audit record.

Configuration / operation	Map groups (history always retained)	Retained compositions	K
B / base	None	None	2
A / all	road, river, cover, surface	All four	28
$A-$ road / LOO	river, cover, surface	River distance-direction; surface-history	19
$A-$ river / LOO	road, cover, surface	Road distance-direction; cover-road; surface-history	23
$A-$ cover / LOO	road, river, surface	Road/river distance-direction; surface-history	20
$A-$ surface / LOO	road, river, cover	Road/river distance-direction; cover-road	20
$B+$ road / LIO	road	Road distance-direction	7
$B+$ river / LIO	river	River distance-direction	7
$B+$ cover / LIO	cover	None (road absent)	6
$B+$ surface / LIO	surface	Surface-history (history retained)	10

The shared B_0 is fitted first by ridge regression with strength 10^{-6}. Each of the 404 outer-training windows contributes one actual first-future transition $(\Delta x_i, \Delta t_i)$; the elapsed times need not equal the method interval τ. Then configuration g fits the remaining rate $y_i = \Delta x_i/\Delta t_i - B_0^\top r(Z_i)$ by

$$\widehat{\theta}_g = \operatorname*{argmin}_\theta \left\{ \frac{1}{2n}\|D_g\theta - Y\|_F^2 + \frac{\lambda}{2}\|\theta\|_F^2 \right\}, \quad n = 404, \quad \lambda = 10^{-4}. \tag{14}$$

D_g stacks the augmented feature rows. This two-stage fit is linear in its parameters; map lookup and history feedback nevertheless make its position dynamics state dependent. Each configuration estimates its own diffusion from fitted displacement residuals, $Q_g = L_g L_g^\top = n^{-1}\sum_i(\Delta x_i - \widehat{b}_{g,i}\Delta t_i)(\Delta x_i - \widehat{b}_{g,i}\Delta t_i)^\top/\Delta t_i$. The diffusion estimate differs from the method branch's centred τR. Euler–Maruyama advances each path with drift $b_g\Delta$ and noise $\sqrt{\Delta}L_g z$; every five saved seconds the history buffer appends its generated position and recomputes the secant. Maps are queried at the predicted position. Future observed positions never enter this buffer.

The base configuration retains the shared history drift and its fitted history-only residual conditioner; it queries no map layer. All-terrain adds all registered groups and interactions; LOO removes one group from that full configuration, and LIO adds one group to the history baseline. Thus RQ2 compares these forecasting configurations on matched targets. The shared baseline and separate residual fits make the intended increment explicit, while refitting and feature penalties limit attribution to an individual factor.

Map sources and attribution limits. Model-source declarations identify WorldCover 2021 v200 [16], Overture transportation 2026-08-19.0 [23, 17], HydroRIVERS v1.0 [18, 19], and retained Mapzen Skadi / Copernicus GLO-30 rules [20, 21, 22]. These product declarations do not certify tile/track contemporaneity. Road/river direction points to the nearest feature, not along its tangent or flow; the retained roads are not the complete pedestrian network.

Constant validity masks, penalised intercepts, unstandardised features and refitted diffusion

make effective regularisation differ among configurations. Thus even a complete comparison would concern configuration increments, not isolated terrain information. The scientific supplement gives full equations, input geometry, excluded road classes, fitting rules and procedure-specific controls. The terrain research question remains separate from ordinary HSM.

3 Evaluation measures and inference

For particles $x_1, \ldots, x_N$ and target y at one matched horizon, the execution-bound finite-ensemble energy score is

$$\mathrm{ES}_N = \frac{1}{N}\sum_i \|x_i - y\| - \frac{1}{2N(N-1)}\sum_{i\neq j}\|x_i - x_j\|, \qquad N \geq 2. \tag{15}$$

The second sum uses ordered pairs. For the deterministic CV reference, ES is directly $\|x - y\|$. ES is measured in metres and lower is better; weighted ES is the equally weighted average over the four targets. Mean-position error is $e_j = \|\bar{x}_j - y_j\|$, with ADE $= \frac{1}{4}\sum_j e_j$ and FDE $= e_4$. FDE is neither best-particle nor average-particle error, and ADE is not a dense-path integral. ES includes interparticle spread and cannot be converted into error of the predictive mean. The first ES term measures particle distances to the target; the second subtracts half the average interparticle distance. Their balance evaluates the distribution, rather than the distance from its centre alone. Consequently, changing spread or mixture geometry can change ES without moving the mean closer to the target. This motivates reporting both quantities rather than using ES as a proxy for localisation.

What a prediction disk promises. At each future target time, the model generates 512 possible positions. It draws a circle around their mean containing about 90% of those simulated positions: this is the nominal 90% prediction disk. That 90% is initially a model-internal proportion, not proof that the model is right. Empirical coverage instead asks how often the actual target falls inside the disk. A reliable 90% disk should contain actual targets about 90% of the time across comparable repeated forecasts.

More generally, a nominal level-q disk is centred at the ensemble mean, with radius the $\lceil qN \rceil$-th ordered distance of particles from that mean. Coverage records whether the true target lies inside; area is πr^2. Mean area is $\mathrm{E}[\pi r^2]$, not $\pi \mathrm{E}[r]^2$. These are horizon-wise marginal regions, not joint path regions. Enclosing about 90% of simulated particles does not guarantee 90% coverage of real targets. For multimodal clouds, the mean may lie in a low-density gap and a disk may include inaccessible terrain; it is not a highest-density or route-constrained region. For $\widehat{P}_j$'s own disk $D_{q,j}$, nominal mass concerns $\widehat{P}_j(D_{q,j}) \approx q$. Empirical reliability instead concerns the observed frequency of $Y_j \in D_{q,j}$ across origins. The two coincide only under suitable calibration of the predictive distribution; the particle construction itself supplies no such guarantee. Area records how much spatial information that coverage costs.

Mean-position error: How far is the predicted centre from the actual target?

Prediction-disk coverage: Does the stated region actually contain the target, and how often?

Prediction-disk area: How much space must be included to obtain that containment?

Coverage alone is insufficient: a disk as large as an entire city may contain more targets while offering little useful localisation. Nor does a miss prove that the radius alone is too small; an inaccurate centre can also put the target outside the circle.

For each recording block, the five seed values are averaged first; blocks then receive equal weight. There are 46 targets per horizon, not 230 independent targets. Six known-velocity and six point-only cases reuse primary records, giving 58 origin–mode cases rather than 58 independent blocks. Original family-wise simultaneous bootstrap-t intervals use 2,000 draws per stream, with shared block indices within a family; Holm-adjusted zero tests have a separate role [13, 14]. Intervals condition on the saved fits and unverified block independence / exchangeability; seeds represent simulation variation, not training or participant replication.

Deltas are candidate minus the reference procedure specified for that comparison. The registered practical margin is $\delta = 41.100259$ m, 5% of validation CV ES, not a validated operational utility threshold. Practical benefit/harm requires the entire original interval beyond $-\delta/+\delta$ and the original statistical, seed and qualification gates. Crossing zero is not equivalence. Practical equivalence, inconclusive information and unavailable evidence are distinct. The scientific supplement retains precise endpoints and finite bootstrap-tail limitations. The descriptive horizon and coverage summaries below are not new superiority or calibration hypothesis tests.

4 Results

4.1 CV is better at 1 and 5 minutes; HSM-SDE at 15 and 30

In this evaluation, CV performs better at 1 and 5 saved minutes, and HSM-SDE at 15 and 30; here "better" means lower average ES across the 46 recording blocks, not an improvement on every metric or every record. HSM-SDE's weighted ES is 477.85 m versus 824.73 m for CV; the saved difference is -346.87 m, computed before rounding. At 1 minute, HSM-SDE/CV ES is 50.57/31.17 m; at 5 minutes it is 243.23/237.93 m. Thus CV has lower ES at 1 and 5 minutes and HSM-SDE at 15 and 30 minutes (Figure 2). The same figure separates ES from mean-position error. Four discrete targets do not identify an exact crossover time. The result supports conditional longer-horizon

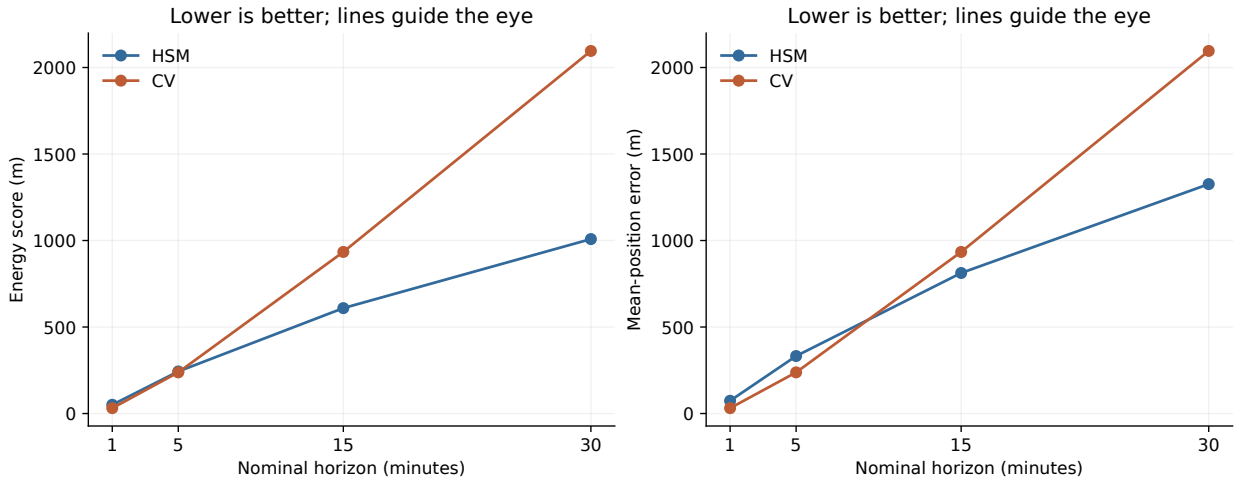


Figure 2: HSM-SDE versus CV at four nominal saved-clock horizons: ES (left) and predictive-mean error (right). Each point averages five seeds within each of 46 primary blocks, then weights blocks equally; inertia is deterministic. Lower is better. Connecting lines guide reading, not interpolation or evidence for a precise crossover.

utility relative to CV, not uniform improvement at every horizon. The deterministic reference is a

Table 4: Saved predictive-mean errors (m), using the original matched targets. Rows use the unified reader-facing procedure names. These are not ES or mean particle errors.

Model	1 min	5 min	15 min	30 min
HSM	73.77	332.31	812.26	1326.09
RDM	71.92	323.79	792.99	1299.11
Obs 300 s	75.88	340.10	825.76	1335.73
CV	31.17	237.93	934.03	2095.78

valid point-mass distribution for ES. It does not establish that HSM-SDE's stochastic complexity is necessary relative to a simple probabilistic constant-velocity or correlated-random-walk model.

4.2 Design choices within the mixture-SDE family

4.2.1 RDM-SDE has better mean ES and final location than HSM-SDE

RDM-SDE's average ES and thirty-minute mean-position error are both lower than those of the main HSM-SDE model (reference code HSM-01). This comparison changes both grouping-based fitting and forecast-time component persistence; it is not a grouping-only controlled comparison. The average ES difference is -2.83 m. Its final mean-position error is also lower: 1299.11 versus 1326.09 m (Table 4). Its mean-position errors are lower at all four saved targets, but that does not establish lower ES at each horizon. All five seed ES differences are negative; the simultaneous block interval, $[-21.22, 16.33]$ m, nevertheless includes zero. RDM-SDE therefore has better observed mean performance, but a consistent ES advantage over HSM-SDE has not been established. The pointwise variant instead increases average ES by 32.95 m with interval $[22.99, 43.37]$ m. The single-Gaussian and explicit-feature variants change ES by $+22.01$ and $+57.96$ m, with intervals including zero. None establishes a registered practical benefit or harm. These comparisons concern fitted procedures, not the general merit of mixtures or velocity information.

4.2.2 Observation interval: lower ES can accompany worse final location

Relative to the 60 s HSM-SDE reference, the 120 and 300 s configurations have lower average ES, but the 300 s configuration has higher thirty-minute mean-position error. Thus its distribution-score improvement comes with a location cost. The 120 s variant changes average ES by -11.56 m, with simultaneous interval $[-20.37, -3.04]$ m; the 300 s variant changes it by -23.40 m. Observation interval and training/adaptation are two separate design factors. Reading their saved mean changes is not a direct statistical comparison of the two design classes or a causal ranking of their importance. The interval is a fitting-data and update-cadence choice, not the integration step. Its effect must also be judged against mean-position error: The 300 s observation-interval procedure illustrates the distinction. Relative to the main HSM-SDE model (reference code HSM-01), weighted ES changes by -23.40 m but FDE by $+9.64$ m. Its mean-position error is also larger at each of the four saved targets (Table 4). ES accounts both for distances to the target and for spread between particles; it cannot be converted into mean-position error. The observation-interval comparison changes training exposure and diffusion binding as well as cadence, so this contrast does not isolate a numerical integration effect.

The dt300 ES simultaneous interval is $[-48.27, -2.47]$ m. Excluding zero does not satisfy the registered practical-benefit criterion: the whole interval would have to be below -41.100259 m,

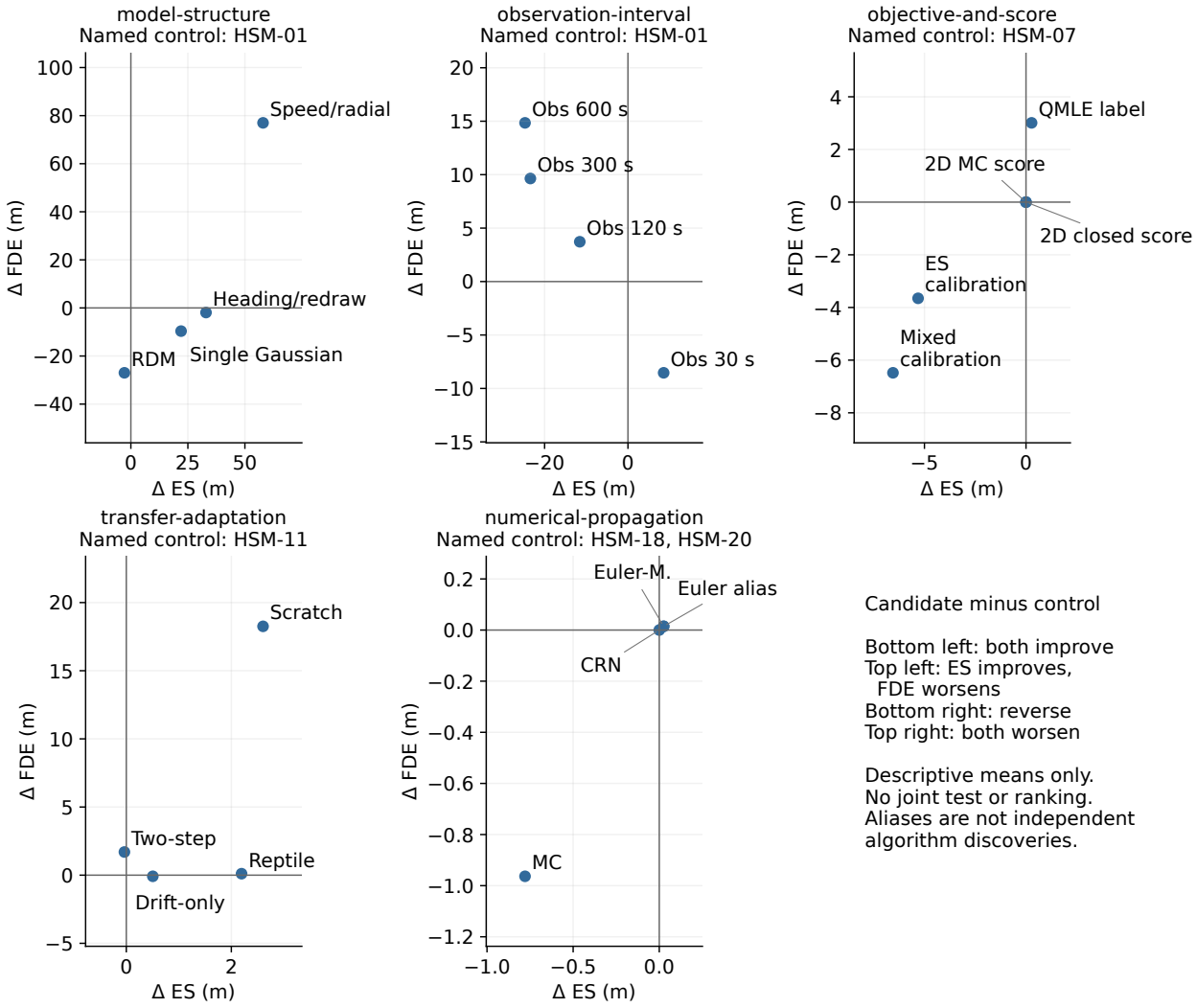


Figure 3: Distribution-score versus endpoint-error trade-offs. All 21 comparisons (42 displayed scalar differences) retain five registered families and named controls. Each point averages five simulation seeds within each of 46 paired blocks, then weights blocks equally; ES first averages four targets while FDE uses only the last. Negative is favourable on both axes; the upper-left quadrant improves ES but worsens FDE. Facets have different scales. These means have no joint confidence region and do not establish a Pareto or global algorithm ranking.

with the other gates met. The point thus communicates a metric conflict, not a practically superior method. All differences are calculated from unrounded source values.

4.2.3 Training and adaptation: no confirmed additional ES decrease

Training initialisation and parameter adaptation are separate choices. The examined variants make only small average ES changes relative to the original training/adaptation procedure; no additional ES decrease is confirmed. The scratch and Reptile variants change average ES by +2.60 and +2.19 m; their intervals are $[-4.73, 10.02]$ and $[-11.95, 12.91]$ m. The drift-only and two-step adaptation change ES by +0.50 and −0.04 m, with intervals $[-1.52, 2.49]$ and $[-2.60, 1.81]$ m. The original training/adaptation procedure is this group's reference (reference code HSM-11), not a replacement for the main model in the horizon comparison. All four original finite-procedure states are practically equivalent within the registered margin. The examined training/adaptation complexity therefore shows no clear added ES benefit; this is not evidence that learning itself is unnecessary. Fixed fitted models and simulation seeds do not measure variation across independently repeated training runs.

4.3 Long-horizon prediction disks are not empirically reliable

Generating a probability distribution and giving a trustworthy region are different achievements. A nominal 90% disk is intended to contain real targets at roughly that frequency, not merely 90% of simulated particles. HSM-SDE's nominal 90% disk coverage is 95.65%, 37.83%, 35.22% and 46.52% at the four horizons. At one minute the nominal 50% disk covers 33.04%, while at thirty minutes even the nominal 95% disk covers only 52.17%. The departure therefore depends on horizon and probability level; no radius rescaling has been tested. Across all 28 identities, thirty-minute 90% coverage ranges from 21.74% to 79.13%. Thus, in this evaluation, the thirty-minute HSM-SDE disks achieve a seed-averaged inclusion rate of only 46.52%, rather than the approximately 90% expected of reliable 90% regions: their nominal containment promise is not met in this evaluation. Figures and tables use HSM and RDM consistently. The dt300 disk is about twice HSM's mean

Table 5: Thirty-minute nominal 90% disks: three configurations and nine saved summary values. Each configuration uses 46 blocks and five ensembles per block. Radius and area are averaged separately; the radius of the mean area is not the mean radius.

Model	Coverage (%)	Mean area (km^2)	Mean radius (m)
HSM-01	46.52	4.534	1183.6
RDM	45.22	3.079	989.5
dt300	64.35	9.109	1692.8

area, yet its inclusion rate remains below the nominal 90%. A larger region can include more targets while its centre is less accurate. Conversely, RDM's smaller region does not by itself establish better uncertainty prediction. Coverage averages the same real target across five simulated disks in each block; there are 46 targets per horizon, not 230 independent targets. At block level this five-seed inclusion proportion takes only 0, 20, 40, 60, 80 or 100%. The 552 points in Supplement Figure 16 can overlap at those six levels; neither a block median of 100% nor a dense-looking cluster is a conditional calibration certificate. These observations support reporting size and inclusion together while retaining the complete block summaries. The 600 s procedure also has a larger thirty-minute mean area (12.609 km^2) and higher coverage (79.13%) than HSM, but its FDE is 1340.94 m. This

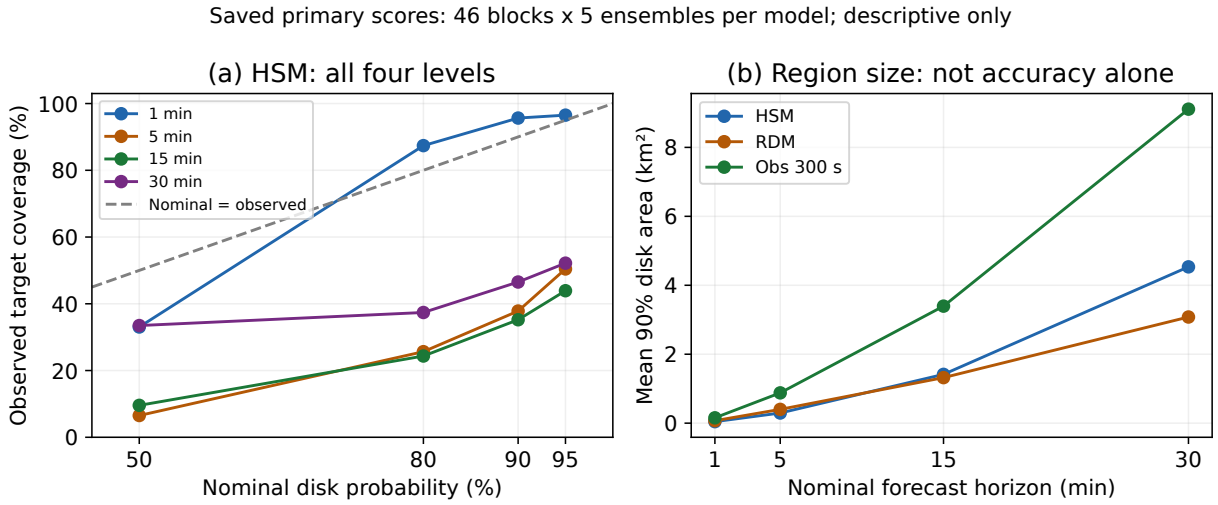


Figure 4: Calibration and size from the saved 46-block, five-seed summaries. Left: 16 HSM coverage values (four probability levels at four horizons); closeness to the diagonal, not the highest curve, is the target. Right: 12 mean areas (three configurations at four horizons), to be read with coverage rather than ranked alone. Connecting lines guide reading; they do not add intermediate horizons or a calibration test.

is another descriptive size–coverage trade-off, not evidence of calibrated uncertainty or improved mean location. No radius or parameter is retuned after inspection of this cohort.

4.4 Qualification of the registered method comparisons

Of the 21 registered primary method contrasts, 14 have practically equivalent and seven inconclusive finite-algorithm states; no original simultaneous interval lies wholly beyond the practical-benefit or harm boundary. All resolution-invariant states remain inconclusive. These states qualify the implemented finite procedures, not general algorithms. For example, all five RDM seed effects are negative but its block interval is $[-21.22, 16.33]$ m. Seed-direction agreement does not resolve uncertainty across recording blocks. Complete effects, seeds and qualification reasons are retained in Supplement Table 25 and the accompanying qualification discussion.

4.5 What the saved terrain predictions show

Question and comparable evidence. The terrain branch asks whether fitted map-conditioned corrections improve on its history-only base, not whether ordinary HSM-SDE beats CV. All ten configurations were fitted and forecasts were executed. The original primary grid contains 2,300 distinct slots: 2,297 successful and three failed, with no pending or missing slots. To compare the same observations, we describe the intersection of 43 recording blocks for which all ten configurations have all five successful seed forecasts (2,150 slots). Three entire blocks are excluded from this post-hoc comparison; no failed result is set to zero or replaced. Selection uses execution status, not which model wins, but still conditions on successful execution. These summaries do not estimate performance on the original full 46-block population or replace its registered inference. Seeds are averaged within a block before blocks receive equal weight. All underlying scores and forecasts are unchanged.

Overall combination: no descriptive aggregate gain. The current all-terrain configuration is worse than its history base on both average ES and thirty-minute mean-position error. In this implementation, simply adding every terrain feature has not produced an average gain. On this common subset, all-terrain has average ES 482.06 m versus 455.08 m for base, a change of +26.98 m, and FDE 1459.37 versus 1420.37 m, a change of +39.00 m (Table 6). Its ES is higher at all four targets; the 30-minute difference is +69.11 m (Figure 5). Yet all-terrain has lower average ES in 22 of the 43 individual blocks. Thus the aggregate disadvantage coexists with heterogeneous local gains; selected successful case maps cannot establish an average terrain benefit. This is a descriptive result about this fitted combination on the selected subset, not a significant harm verdict or proof that terrain information is intrinsically unhelpful.

Table 6: Terrain predictions on the same 43 complete recording blocks (five seeds each). ES averages four targets; ΔES is relative to B (history). FDE, coverage and area use 30 minutes. Success counts retain all 230 original slots per configuration. These are post-hoc descriptive means, not the registered full-family effects.

Configuration	ES (m)	ΔES (m)	FDE (m)	Cover (%)	Area (km^2)	Success
B (history)	455.08	+0.00	1420.37	46.05	5.503	230/230
A (all groups)	482.06	+26.98	1459.37	40.47	5.609	229/230
B + road	498.04	+42.96	1493.33	52.09	6.202	229/230
B + river	455.27	+0.19	1478.38	41.40	5.701	229/230
B + cover	453.44	-1.64	1414.40	49.77	5.635	230/230
B + surface	462.66	+7.58	1410.27	47.44	5.670	230/230
A - road	462.19	+7.11	1459.69	44.65	5.719	230/230
A - river	498.29	+43.20	1446.22	50.23	5.693	230/230
A - cover	482.14	+27.06	1465.55	41.40	5.886	230/230
A - surface	477.11	+22.03	1485.96	46.51	6.145	230/230

Factor configurations: context and horizon matter. Adding the current road features accompanies worse scores; removing them improves scores in the compared configurations. Adding the road group alone increases average ES by 42.96 m relative to base; removing it from all-terrain decreases ES by 19.87 m. Both directions associate the current road representation with a score cost in these fitted configurations, not a general claim that roads cannot help prediction. The river-only configuration's benefit does not carry from 15 to 30 minutes: improvement at a shorter horizon does not imply improvement farther ahead. It reduces 15-minute ES by 42.51 m but increases 30-minute ES by 57.27 m relative to base. Its average ES change is only +0.19 m, whereas removing river from the full combination increases average ES by 16.22 m. A factor's standalone result therefore does not determine its role in the full configuration.

Adding land cover alone lowers average ES by 1.64 m and final mean-position error by 5.96 m, showing a small favourable mean result. Its average ES is 453.44 m, the lowest of these ten saved configuration means, not a confirmed general advantage. Surface-only instead increases average ES by 7.58 m while lowering FDE by 10.09 m. These illustrate a small favourable configuration and a score–location trade-off, respectively, not confirmed factor superiority. Fits, diffusion, interactions and effective regularisation change across configurations, so these differences cannot be assigned to isolated causal terrain effects.

Larger paired subsets and prediction-region limits. As a descriptive sensitivity check, each candidate/control pair can retain all of its five-seed-complete blocks: 45 for comparisons involving

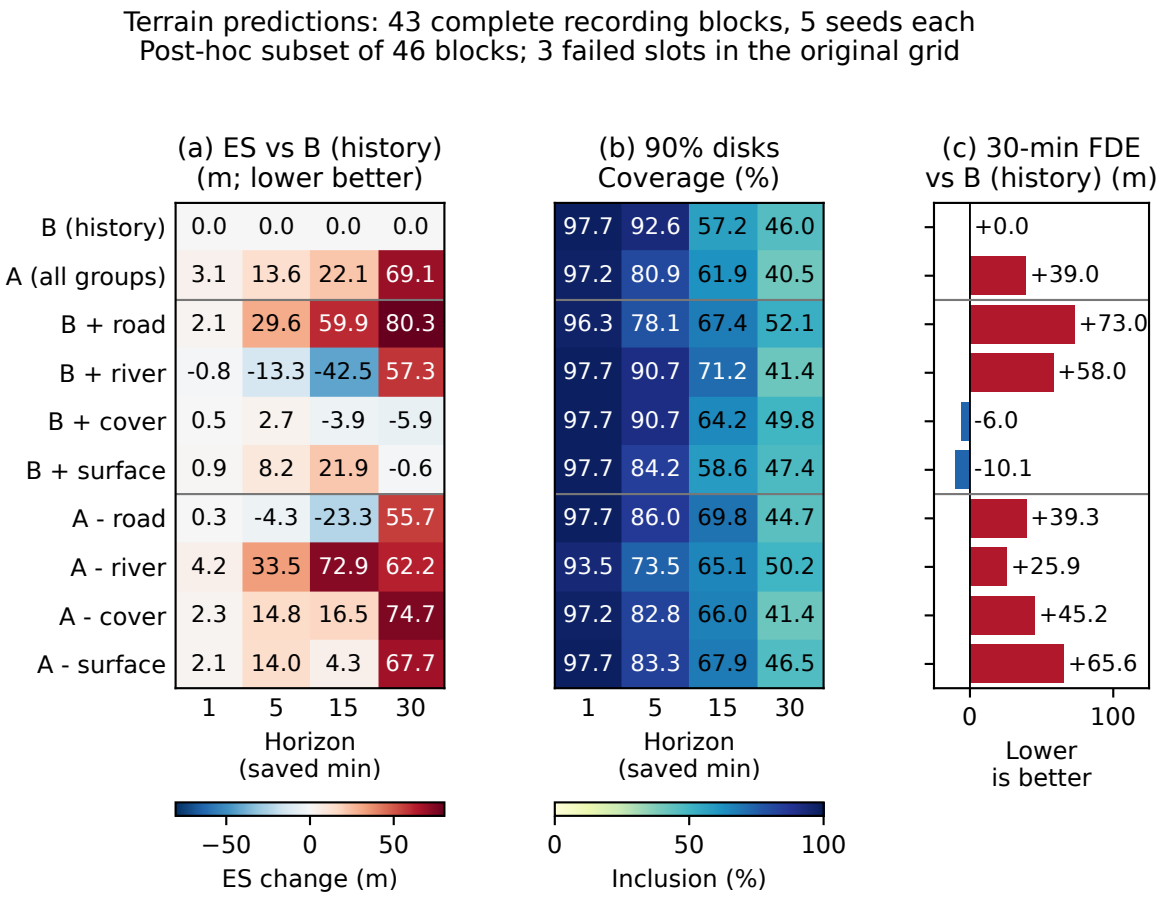


Figure 5: Existing terrain forecasts on the same post-hoc 43-block subset. Left: four horizon-specific ES differences from history-only base; negative is favourable. Middle: actual inclusion in nominal 90% disks, with 90% as the calibration target. Right: 30-minute predictive-mean error differences from base. All panels use identical blocks and equal five-seed/block weighting. LIO adds one group to base; LOO removes a group and dependent interactions from all-terrain. Panels compare configuration performance, not isolated factor effects. Missingness in the full grid is retained in Table 6; no inferential intervals are supplied.

a failed configuration, 46 for complete single-group/base pairs. All-terrain/base then changes ES by +21.17 m; road-only/base by +41.18 m; removing road by −18.50 m; and land-cover-only/base by −3.81 m. Those directions agree with the common-subset summaries. River-only/base changes from +0.19 to −1.54 m, showing the fragility of its near-zero average. The larger subsets also condition on success, use different blocks and cannot repair the missing full-cohort estimand or justify a global ranking.

Thirty-minute nominal 90% coverage spans only 40.47–52.09% across the ten configurations on the common subset. All-terrain covers 40.47% versus base's 46.05%, despite slightly greater mean area, 5.609 versus 5.503 km^2. Terrain conditioning has therefore not made the long-horizon disks empirically reliable in this comparison. Taken together, the saved data show configuration-dependent and horizon-dependent trade-offs, not a uniform gain from adding every terrain group. The registered full-family answer remains separately unavailable under the original rule below.

Three required primary terrain forecasts failed: all-terrain once and LIO-river/LIO-road once each. LOO means leave-one-group-out and LIO means leave-one-group-in. The overall/LOO grid has 1379/1380 successes and LIO has 1148/1150, with no pending forecasts; the grids share baselines and are not additive. Under the original complete-family rule, both whole families are unavailable. Successful subsets or individual map cases cannot replace that rule. This is not a positive or null terrain result. Two further failures affect auxiliary known-velocity method families, not the complete primary method families. Failure identities and logging limits are in the audit record; missing exception stacks prevent attributing a specific technical cause.

5 Discussion

5.1 What the model description lets us interpret

The task definition connects the horizon result to the two reference constructions. CV extends a short observed secant as a single straight trajectory; it can remain accurate when that local rate persists, but any rate error is multiplied by elapsed time. HSM-SDE instead assigns probability to several fitted drift/diffusion components, without updating their weights from the new prefix's heading. This gives different uses of recent movement and different uncertainty representations. The observed short/long ordering is consistent with a changing forecasting trade-off, but does not identify which mechanism causes it: the comparison changes both mean dynamics and distributional spread, and no task-matched probabilistic inertia control isolates these effects. The constant-drift covariance explanation above is not a fitted explanation of the observed reversal.

The other two diagnostics follow from the forecast summaries. ES evaluates both distances to the realized target and distances within the ensemble, whereas FDE evaluates only the final mean. An ES decrease with FDE increase therefore establishes a disagreement between criteria, not a mathematical contradiction. A radial particle quantile describes the model's own spread; coverage compares that region with actual target positions. The observed undercoverage rejects the interpretation of HSM-SDE's nominal disks as empirically reliable regions for this cohort, even when average ES is favourable. It does not identify whether drift error, insufficient spread, mixture geometry or selection is responsible.

5.2 Which design choices merit attention

The observed long-horizon benefit makes the mixture-SDE family worth examining beyond an average all-horizon score. Within it, RDM-SDE provides a favourable descriptive structural result, whereas the training/adaptation variants provide no confirmed additional ES decrease.

Observation-interval choices can lower ES while worsening final mean-position error. These observations prioritise design questions, not a global algorithm ranking: the contrasts use their own controls, some change several factors, and the original practical/inference qualifications still apply.

The objective-choice comparisons also require care. HSM and QMLE retain identical saved drift, covariance and mixture weights; mixed-loss and pure-ES variants likewise retain identical saved parameters to each other. Differences between those paired identities' finite scores do not prove distinct learned predictive models or the superiority of one training loss.

5.3 Reliability limits and remaining work

The pipeline can fit and generate probabilistic forecasts and score them at matched targets. The horizon result supplies conditional utility evidence relative to a deterministic reference. Neither establishes a reliable long-horizon prediction region: that requires empirical coverage at the stated level while retaining informative area and location. The coverage findings limit this capability even where average ES improves.

The calibration proxy differs from the evaluated predictive distribution, and RDM training/validation component labels differ. Both are plausible contributors to poor uncertainty behaviour, but these results do not identify causes or validate a correction. Likewise, wider regions can increase inclusion while retaining an inaccurate centre; coverage alone would reward sufficiently uninformative regions. The metric conflict is therefore substantive, not a reason to discard an unfavourable metric.

The scope is deliberately conditional. Retrospective future-route selection, previous evaluation exposure, uncertain recording independence and unverified clock provenance limit population and physical-time interpretation. Fixed fits and seeds omit training uncertainty. Parameter aliases, unmatched mode labels, joint procedure changes and representation-dependent penalties limit component attribution. The missing matched probabilistic baseline precludes a claim that complex stochastic structure is necessary. None of these limitations is removed by reproducing saved numerical summaries.

The descriptive terrain analysis gives a more specific development target. The current all-terrain configuration is worse than its history base on both average ES and thirty-minute mean-position error; simply stacking terrain features has not produced an average gain here. Road addition accompanies worse scores and removal improves them; the river-only gain at 15 minutes does not extend to 30. Land-cover-only shows a small favourable mean, not an established general benefit of terrain information. These findings raise a substantive possibility that the current handling of terrain is inadequate. The road/river directions point towards nearest feature points, not along roads, and do not explicitly model river barriers and crossings. Such input choices need not encode the movement mechanism one expects. Representation, fitting and long-horizon use are therefore plausible sources of the disappointing results, but this study does not identify or validate a specific cause. This is an interpretation, not an additional empirical finding that the factors are intrinsically noise.

Future work should compare a task-matched probabilistic motion baseline, align mode labels before transfer and calibration, and evaluate calibration of the actual multi-step distribution. Terrain comparisons require complete families and matched effective penalties. Independent route grouping and clock-source verification are needed for broader inference. These are unexecuted studies, not results or editorial repairs supplied here.

6 Conclusion

Which model is better, and when. In this 46-block saved-clock evaluation, CV is better at 1 and 5 minutes; HSM-SDE is better at 15 and 30 minutes. "Better" here means lower average ES.

RDM-SDE versus the main HSM-SDE model. RDM-SDE has lower average ES and thirty-minute mean-position error than the main HSM-SDE model. Its ES difference interval includes zero, so the observed means favour RDM-SDE without establishing a consistent ES advantage.

Observation interval. The 120 and 300 s configurations have lower average ES than the 60 s reference, but the 300 s configuration has higher final mean-position error. Its score improvement therefore comes with a location cost.

Training and adaptation strategies. The examined variants change average ES only slightly relative to the original training/adaptation procedure; no additional ES decrease is confirmed. This concerns changing those strategies, not whether training itself is useful.

Prediction regions. HSM-SDE's thirty-minute nominal 90% disks achieve empirical coverage of only 46.52%, rather than approximately 90%. Their reported containment promise is not met. Mean-position error, coverage and area must be read together: a larger circle can cover more targets without locating them well.

Terrain configurations. On the post-hoc 43-block common subset, the current all-terrain configuration is worse than its history base on both average ES (+26.98 m) and thirty-minute mean-position error (+39.00 m). Simply adding all terrain features has not produced an average gain in this implementation. Adding the current road features accompanies worse scores; removing them improves scores. The river-only configuration's 15-minute gain does not extend to 30 minutes. Adding land cover alone lowers average ES by 1.64 m and final mean-position error by 5.96 m, a small favourable mean result. No configuration's thirty-minute nominal 90% coverage exceeds 52.09% on this subset. The evaluation has not established that terrain information reliably helps forecasting. It raises the possibility of shortcomings in current terrain handling, but does not establish that explanation as a cause.

Inference qualification, not an additional finding. RQ1 yields 14 practically equivalent and seven inconclusive finite-procedure contrasts, with all resolution-invariant judgements inconclusive. RQ2 does not meet the registered complete-family rule because three required terrain forecasts failed; no benefit, harm or equivalence verdict for that family can be claimed under that rule. The terrain descriptions above do not replace it or identify causal factor effects. Carrier identity and clock provenance remain unverified, and retrospective selection and prior evaluation exposure limit generalisation. These qualifications do not erase the observed comparisons, but prevent general superiority or operational guarantees.

Data and reproducibility. The accompanying scientific supplement contains full specifications and all 50 tables; the separate audit record contains execution and revision history. The public repository provides saved aggregate evidence and reproduction code; private trajectories and local case maps are not redistributed. Repository: `https://github.com/HuAzurestar/learnable-sde-paper-to-predict-movement`. The public argument does not rely on those private case

illustrations. Trajectory research-use permissions and the applicable ethics-review / consent basis require author verification; public availability of a source or map product is not taken as evidence of that authorization. This manuscript revision changes no fit, forecast, score or inference rule. Author identities, affiliations and contribution/conflict declarations require owner confirmation before submission and are not inferred from repository accounts.

References

[1] D. S. Johnson, J. M. London, M.-A. Lea and J. W. Durban (2008). Continuous-time correlated random walk model for animal telemetry data. *Ecology*, 89(5), 1208–1215. https://doi.org/10.1890/07-1032.1.

[2] Z. Ghahramani and G. E. Hinton (2000). Variational learning for switching state-space models. *Neural Computation*, 12(4), 831–864. https://doi.org/10.1162/089976600300015619.

[3] T. Avgar, J. R. Potts, M. A. Lewis and M. S. Boyce (2016). Integrated step selection analysis: bridging the gap between resource selection and animal movement. *Methods in Ecology and Evolution*, 7(5), 619–630. https://doi.org/10.1111/2041-210X.12528.

[4] (2017). Corrigendum. [Correction to Avgar et al. (2016)]. *Methods in Ecology and Evolution*, 8(9), 1168. https://doi.org/10.1111/2041-210X.12725.

[5] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese and A. Alahi (2018). Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2255–2264. https://openaccess.thecvf.com/content_cvpr_2018/html/Gupta_Social_GAN_Socially_CVPR_2018_paper.html.

[6] T. Salzmann, B. Ivanovic, P. Chakravarty and M. Pavone (2020). Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, 12363, 683–700. Springer. https://doi.org/10.1007/978-3-030-58523-5_40.

[7] P. Kidger, J. Foster, X. Li and T. J. Lyons (2021). Neural SDEs as Infinite-Dimensional GANs. *Proceedings of the 38th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 139, 5453–5463. https://proceedings.mlr.press/v139/kidger21b.html.

[8] T.-T. Gao, B. Barzel and G. Yan (2024). Learning interpretable dynamics of stochastic complex systems from experimental data. *Nature Communications*, 15, 6029. https://doi.org/10.1038/s41467-024-50378-x.

[9] T. Gneiting and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. https://doi.org/10.1198/016214506000001437.

[10] T. Gneiting, L. I. Stanberry, E. P. Grimit, L. Held and N. A. Johnson (2008). Assessing probabilistic forecasts of multivariate quantities, with an application to ensemble predictions of surface winds. *TEST*, 17, 211–235. https://doi.org/10.1007/s11749-008-0114-x.

[11] P. E. Kloeden and E. Platen (1992). *Numerical Solution of Stochastic Differential Equations*. Springer. https://doi.org/10.1007/978-3-662-12616-5.

[12] A. Nichol, J. Achiam and J. Schulman (2018). On First-Order Meta-Learning Algorithms. [Preprint], *arXiv:1803.02999v3*. https://arxiv.org/abs/1803.02999v3.

[13] S. Holm (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. https://www.jstor.org/stable/4615733.

[14] A. C. Davison and D. V. Hinkley (1997). *Bootstrap Methods and their Application*. Cambridge University Press. https://doi.org/10.1017/CBO9780511802843.

[15] NOAA Global Monitoring Division (n.d.). *General Solar Position Calculations*. [Online documentation]. Accessed 7 October 2026. https://gml.noaa.gov/grad/solcalc/solareqns.PDF.

[16] D. Zanaga et al. (2022). *ESA WorldCover 10 m 2021 v200*. [Dataset], version v200. https://doi.org/10.5281/zenodo.7254221.

[17] Overture Maps Foundation (n.d.). *Attribution and Licensing*. [Online documentation]. Accessed 6 October 2026. https://docs.overturemaps.org/attribution/.

[18] B. Lehner and G. Grill (2013). Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186. https://doi.org/10.1002/hyp.9740.

[19] HydroSHEDS (n.d.). *HydroRIVERS product documentation and licence*. [Online documentation], version v1.0. Accessed 6 October 2026. https://www.hydrosheds.org/products/hydrorivers.

[20] Mapzen / Tilezen (2017). *Terrain Tiles: Types of Terrain Tiles and Skadi format*. [Dataset distribution documentation]. Document revision e3d4351; accessed 7 October 2026. https://registry.opendata.aws/terrain-tiles/. https://github.com/tilezen/joerd/blob/e3d4351ee3e5be333e23f47cb500e9a7c310656a/docs/formats.md.

[21] Mapzen / Tilezen (2017). *Terrain Tiles: Attribution*. [Online documentation]. Document revision d8f587b; accessed 7 October 2026. https://github.com/tilezen/joerd/blob/d8f587b73d26e0a0c42cdccd9ae8c55de4197763/docs/attribution.md.

[22] Copernicus Programme / Sinergise (2021). *Copernicus Digital Elevation Model: GLO-30 Public*. [Dataset distribution documentation], 2021 release; accessed 7 October 2026. https://registry.opendata.aws/copernicus-dem/.

[23] Overture Maps Foundation (2026). *2026-08-19 release notes*. [Dataset release documentation]. Data version 2026-08-19.0; accessed 7 October 2026. https://docs.overturemaps.org/blog/2026/08/19/release-notes/.